A RESEARCH PAPER OF THE SERIES ON INTEGRATED AUTONOMOUS INTELLIGENCE

NAVIGATION-CONDITIONED KNOWLEDGE EVOLUTION

How Search Policies Shape Reasoning, Memory and Institutional Intelligence in LLM–Knowledge Graph Systems

COGNITIVE TOPOLOGY · NB13–NB16

"The knowledge available to a model is not the knowledge stored in the graph. It is the knowledge made active by a path through the graph."

ALEJANDRO REYNOSO

JUDGE BUSINESS SCHOOL · UNIVERSITY OF CAMBRIDGE · JULY 2026

Abstract

A knowledge graph gives a large language model access to relationships that a flat document store cannot represent directly. Yet graph construction does not determine graph use. For any non-trivial query, the graph contains more nodes, paths, communities, time layers, and competing explanations than can fit within the model's context window. Some procedure must choose where to begin, which edges to follow, what evidence to exclude, when to stop, and how to allocate a finite retrieval budget. This paper names that procedure the *navigation policy* and argues that it is a constitutive component of the cognitive system, not an implementation detail.

The argument has two levels. At the static level, the same graph, model, query, prompt, and token budget can produce different answers when the navigation policy changes, because each policy constructs a different cognitively active subgraph. At the dynamic level, if governed outputs are incorporated into the vault, the policy also affects which new nodes and edges are proposed, validated, and made available to future queries. Repeated navigation can therefore generate endogenous centrality, hub dominance, epistemic lock-in, confirmation cascades, or, under exploratory policies, excessive novelty and validation burden. Initially identical knowledge systems may diverge solely because they walk through their shared memory differently.

The paper formalizes this process as a coupled state transition over graph topology, navigation, model generation, and governance. It distinguishes the stored graph from the visited graph, the retrieved context, the answer, and the authoritative write-back. It develops a taxonomy of semantic, local, diffusion, community, typed-path, goal-directed, contrastive, exploratory, temporal, and hybrid policies; derives propositions concerning policy non-neutrality, retrieval-induced preferential attachment, and the insufficiency of answer-level provenance; and introduces the concept of a *navigation record* as a canonical governance object.

The contribution is positioned relative to retrieval-augmented generation, GraphRAG, Think-on-Graph, HippoRAG, G-Retriever, adaptive graph traversal, active knowledge navigation, and recent co-evolutionary graph systems. The distinctive focus is not graph navigation alone, nor autonomous graph editing alone, but the controlled identification and governance of navigation-conditioned institutional memory. A four-notebook empirical programme is specified to test static answer divergence, recursive topological divergence, robustness to shocks, and risk-sensitive policy selection. The resulting framework reframes institutional intelligence: the central question is no longer only whether the organization has preserved the right knowledge, but whether it can reach that knowledge through plural, explainable, auditable, and governable paths.

Keywords: knowledge graphs; large language models; GraphRAG; graph navigation; institutional memory; path dependence; retrieval governance; ExoBrain; Obsidian; recursive knowledge systems.

Central proposition

In an LLM–knowledge graph system, navigation is not a neutral retrieval operation. It selects the subgraph that becomes cognitively active, conditions the answer produced, and, when validated outputs are written back, helps determine the topology of future institutional memory.

Series note. This paper advances the ExoBrain research programme from knowledge architecture to cognitive topology. It treats an Obsidian vault as a practical instance of a governed, versioned, typed knowledge graph, while developing a theory independent of any single software environment. Its propositions and hypotheses support the accompanying notebook sequence NB13–NB16.

Executive Orientation

The construction of a knowledge graph is often treated as the difficult part of organizational memory. Documents must be cleaned; entities must be resolved; relations must be typed; dates, sources, and confidence must be recorded; contradictory claims must remain distinguishable; and human authority must be separated from machine-generated candidates. Once this architecture has been built, it is tempting to imagine that the model can simply "use the graph."

That phrase conceals the next problem. A large graph cannot be placed whole inside a finite context window, and a query rarely identifies its relevant subgraph uniquely. The model requires a route. A route may privilege similarity, proximity, centrality, community membership, causal types, decision history, recent evidence, contradictory evidence, or distant weak ties. Each route illuminates some knowledge and leaves other knowledge dark.

This creates a conceptual shift:

$$\boxed{\text{Cognitive system at time } t = \text{stored graph} + \text{navigation policy} + \text{model} + \text{governance}} \tag{1}$$

The graph is the possibility space. Navigation determines the realized evidentiary path through that space. The context shown to the model is a policy-conditioned projection of the graph, not the graph itself.

The issue becomes more consequential when answers are allowed to influence future memory. Suppose the system reviews two companies, traverses a chain linking interest rates to refinancing pressure, and proposes that their combination may reduce a shared vulnerability. If that proposal is validated and written back, future queries may discover the companies through the new relation. But another policy might have traversed competitive overlap, customer concentration, or legal exposure and proposed a different link. The future vault depends jointly on what the vault contained and on how the model moved through it.

This paper develops a theory for that feedback loop. Its principal practical implication is clear: organizations should govern not only *what* a model writes, but *how it arrived at the context from which it wrote*. The route itself must become an auditable object.

Contents

1 From Knowledge Architecture to Cognitive Topology

1.1 The achievement and the missing layer

An Obsidian vault can be more than a collection of notes. With stable identifiers, typed links, metadata, source references, temporal fields, decision objects, and explicit status labels, it becomes a human-readable knowledge graph. It can represent companies, people, instruments, events, recommendations, assumptions, decisions, outcomes, contradictions, and the evidence connecting them. This is a substantial achievement because it externalizes institutional memory in a form that is both navigable by people and computable by machines.

Yet representation is not cognition. A graph specifies what can be reached, but not what will be reached. It encodes possible walks, not the walk selected for a particular task. In a sparse toy graph the distinction may appear trivial. In a mature institutional vault it is decisive: thousands of nodes can be connected through multiple relation types, time periods, levels of authority, and communities of practice.

The relevant question is therefore not:

> Does the vault contain the knowledge required to answer the question?

It is:

> Under a finite budget, which navigation policy makes the right knowledge cognitively active, preserves the relevant alternatives, and provides an auditable basis for action?

1.2 Stored knowledge and active context

Let the vault at time t be a typed, attributed, temporal graph:

$$\mathcal{G}_t = (V_t, E_t, \tau_V, \tau_E, X_t, S_t, \Gamma_t), \tag{2}$$

where V_t and E_t are nodes and edges; τ_V and τ_E are node and relation types; X_t contains attributes; S_t contains source and provenance objects; and Γ_t contains governance status, permissions, and authority labels.

A query q_t does not expose all of $\mathcal{G}_t$ to the model. A navigation procedure $\mathcal{N}$ applies policy π_t under budget B_t:

$$C_t = \mathcal{N}(\mathcal{G}_t, q_t; \pi_t, B_t, \Omega_t), \tag{3}$$

where Ω_t contains access controls, stopping rules, and system constraints. The resulting C_t is the *cognitively active subgraph*: the nodes, edges, source excerpts, summaries, and metadata actually supplied to the language model.

The answer is:

$$\mathcal{A}_t = \mathcal{L}(q_t, C_t; \theta, p_t), \tag{4}$$

where θ denotes the model and p_t the generation prompt and settings. If the model and prompt are fixed, policy-induced variation in C_t remains sufficient to induce variation in $\mathcal{A}_t$.

Definition 1.1 (Cognitive topology). *Cognitive topology is the effective structure of the knowledge space as experienced by a model under a specified navigation policy, query, budget, and access regime. It is not identical to the stored topology of the graph.*

Definition 1.2 (Navigation policy). *A navigation policy is a rule, algorithm, or model-mediated strategy that determines seed selection, eligible transitions, path ranking, budget allocation, branch expansion, evidence retention, and stopping.*

1.3 The operative architecture

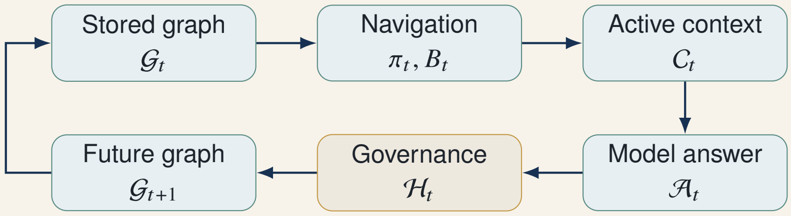


Figure 1. The navigation-conditioned knowledge cycle. The model never receives the stored graph directly; it receives a policy-conditioned active context.

Figure 1 introduces an intermediate layer that is too often compressed into the word "retrieval." That layer can be deterministic, stochastic, model-directed, or hybrid. It may execute a fixed top-k search, a k-hop expansion, Personalized PageRank, community selection, beam search, a typed query plan, or an agentic sequence of actions. Whatever its implementation, it selects the evidence environment in which generation occurs.

1.4 Why this is a theoretical issue

If navigation affected only latency, it would be an engineering problem. If it affected only one answer, it would be an evaluation problem. But when the system incorporates validated outputs into external memory, navigation affects a state transition:

$$\mathcal{G}_{t+1} = \mathcal{G}_t \oplus \mathcal{W}(\mathcal{H}_t[\mathcal{A}_t, C_t, \pi_t]) . \tag{5}$$

The operator $\mathcal{H}_t$ does not merely approve or reject prose. It evaluates candidate knowledge objects, supporting evidence, contradiction status, permissions, and proposed graph edits. The write operator $\mathcal{W}$ converts approved candidates into versioned additions, revisions, qualifications, or deprecations.

Thus:

$$\boxed{\mathcal{G}_{t+1} = F(\mathcal{G}_t, q_t, \pi_t, B_t, \theta, p_t, \mathcal{H}_t)} \tag{6}$$

The future graph is conditioned by its preceding topology and by the process used to navigate that topology. Navigation becomes an endogenous force in the evolution of institutional knowledge.

2 Intellectual Foundations and the Research Gap

2.1 From retrieval to structured retrieval

Retrieval-augmented generation established a general architecture in which a model conditions generation on non-parametric external evidence (Lewis et al., 2020). Conventional dense retrieval typically represents the corpus as passages and ranks them by query similarity. This improves factual grounding and updateability, but it tends to flatten relations that are explicit in organizational knowledge: ownership, causality, chronology, dependency, contradiction, authorization, and decision lineage.

Graph-based retrieval responds by making structure operational. GraphRAG uses an LLM-derived entity graph, community detection, and hierarchical summaries to answer global sensemaking questions that local passage retrieval handles poorly (Edge et al., 2024). G-Retriever formulates relevant subgraph extraction as an optimization problem and connects the selected graph to language generation (He et al., 2024). HippoRAG combines knowledge graphs with Personalized PageRank to support associative, multi-hop memory (Gutiérrez et al., 2024, 2025).

These approaches establish that graph structure can materially improve evidence localization and integration. They also make visible a large design space: local versus global retrieval, centrality-based diffusion, graph optimization, community summaries, and mixtures of graph and vector retrieval.

2.2 From retrieval to navigation

Think-on-Graph treats the language model as an agent that iteratively explores entities and relations using beam search (Sun et al., 2023). PolyG observes that different question types require different traversal strategies and introduces a query planner for adaptive graph traversal (Liu et al., 2025). NaviRAG explicitly frames evidence acquisition as staged navigation inspired by information foraging: first locate a semantic region, then move through it under local signals (Dai et al., 2026; Pirolli and Card, 1999). Adaptive routing work further shows that graph retrieval need not be invoked uniformly; simple questions may be better served by dense retrieval, while complex questions justify graph traversal (Dong et al., 2026).

The central lesson is that no single traversal policy is universally efficient or epistemically adequate. Query structure, task risk, graph quality, and evidence distribution all matter.

2.3 From static graphs to co-evolving graphs

Recent systems move beyond fixed knowledge stores. Agentic-KGR models co-evolution between language models and knowledge graph construction through multi-round reinforcement learning (Li et al., 2025). EvoGraph-R1 treats a hypergraph as an interactive environment in which an agent can retrieve, search, edit, and answer; graph state and policy evolve together (Lin et al., 2026). These systems are important because they dissolve the artificial separation between graph use and graph construction.

They also heighten the governance problem. A reward optimized for answer accuracy or structural performance does not by itself establish organizational authority, evidentiary sufficiency, permission to change canonical memory, or preservation of dissent. In a scientific benchmark, an autonomous graph edit may be an action in an optimization loop. In a bank, regulator, law firm, board, or investment committee, the same edit may alter the future evidentiary environment of consequential decisions.

2.4 The missing object of analysis

Existing work provides strong accounts of:

- how to retrieve external evidence;

- how graph structure can support local, global, and multi-hop reasoning;
- how an agent can choose traversal actions;
- how retrieval policies can be adapted to question types;
- and how graphs and agents can co-evolve.

The missing object is the *governed, longitudinal causal effect of navigation policy on institutional memory*. Four distinctions define the gap.

1. **Answer quality versus memory evolution.** A traversal policy may improve a benchmark answer while producing undesirable long-run concentration, erasure of dissent, or authority drift.
2. **Autonomous editing versus institutional authorization.** The model's ability to propose or execute a graph edit does not make the model an epistemic authority.
3. **Evidence citation versus navigation provenance.** Citing retrieved sources does not reveal the search space, rejected branches, stopping rule, or counter-evidence not visited.
4. **One-policy performance versus counterfactual identification.** To estimate the effect of navigation, the initial graph, question sequence, model, prompt, budget, and governance criteria must be held constant while policy varies.

Research contribution

This paper studies navigation as an epistemic treatment applied to a graph. It asks how alternative treatments change active context, present answers, candidate write-backs, and the future topology of authoritative institutional memory.

3 A Formal Model of Navigation-Conditioned Evolution

3.1 Five objects that must not be conflated

The framework separates five graph-related objects:

1. $\mathcal{G}_t$: the complete stored and access-permitted graph at the start of a query;
2. $\mathcal{V}_t^\pi$: the visited trace, including nodes and edges inspected during navigation;
3. C_t^π: the retained context actually supplied to the model;
4. $\Delta\widetilde{\mathcal{G}}_t^\pi$: candidate graph changes inferred or proposed from the answer;
5. $\Delta\mathcal{G}_t^\pi$: the authoritative changes that survive governance.

Visited information need not be retained. Retained information need not be cited. Cited information need not justify every inferential step. Proposed relations need not be accepted. Accepted relations may be provisional, scoped, or time-bounded. A trustworthy system preserves these distinctions rather than collapsing them into a single "retrieval result."

3.2 Navigation as a sequential decision rule

Let $s_{t,k}$ denote the navigation state after k steps. It includes the query, current frontier, visited set, accumulated evidence, remaining budget, uncertainty, and policy-specific scores. An action $a_{t,k}$ may:

- choose a seed;
- follow an edge;
- expand a community;
- retrieve a source passage;
- open a temporal predecessor;

- search for a contradiction;
- backtrack;
- retain or discard evidence;
- or stop.

The policy is:

$$a_{t,k} \sim \pi(a \mid s_{t,k}; \phi), \tag{7}$$

where ϕ contains fixed parameters or learned weights. The transition is:

$$s_{t,k+1} = T(s_{t,k}, a_{t,k}, \mathcal{G}_t). \tag{8}$$

Navigation ends at stopping time K_t, constrained by:

$$\mathrm{cost}(\mathcal{V}_t^\pi) \leq B_t. \tag{9}$$

The active context is a compression or selection of the navigation trace:

$$C_t^\pi = P(\mathcal{V}_t^\pi, q_t, B_t), \tag{10}$$

where P is a packing function that determines which evidence fits the model's context window. Navigation and context packing are analytically separate: two systems may visit the same nodes but retain different evidence.

3.3 The answer and write-back process

The model produces an answer distribution:

$$\mathcal{A}_t^\pi \sim \mathcal{L}(\cdot \mid q_t, C_t^\pi, \theta, p_t). \tag{11}$$

An extraction operator converts the answer into candidate knowledge objects:

$$\Delta\widetilde{\mathcal{G}}_t^\pi = X(\mathcal{A}_t^\pi, C_t^\pi, \mathcal{V}_t^\pi). \tag{12}$$

Governance evaluates each candidate z against sources, ontology, permissions, contradiction policy, confidence thresholds, and human authority:

$$h_t(z) \in \{\text{reject, quarantine, provisional, authoritative, deprecated}\}. \tag{13}$$

The next graph is:

$$\mathcal{G}_{t+1}^\pi = \mathcal{G}_t^\pi \oplus \{z \in \Delta\widetilde{\mathcal{G}}_t^\pi : h_t(z) \neq \text{reject}\}, \tag{14}$$

with the operator $\oplus$ preserving versions rather than overwriting history.

3.4 Counterfactual policy effects

For two policies π_i and π_j, define static context divergence:

$$D_C(i, j \mid t) = d_C(C_t^{\pi_i}, C_t^{\pi_j}), \tag{15}$$

answer divergence:

$$D_A(i, j \mid t) = d_A(\mathcal{A}_t^{\pi_i}, \mathcal{A}_t^{\pi_j}), \tag{16}$$

and dynamic graph divergence after T rounds:

$$D_G(i, j \mid T) = d_G(\mathcal{G}_T^{\pi_i}, \mathcal{G}_T^{\pi_j}). \tag{17}$$

The distances need not be a single scalar. Context divergence can include node Jaccard distance, edge-type divergence, community coverage difference, and source authority difference. Answer divergence can include claim disagreement, decision disagreement, evidence overlap, and calibration. Graph divergence can combine graph edit distance, degree distribution, modularity, relation-type entropy, and contradiction retention.

Proposition 3.1 (Navigation non-neutrality)**.** *If two admissible policies produce non-identical active contexts with positive probability, and the model's answer distribution is conditionally sensitive to at least one differing item, then answer distributions will differ with positive probability even when the stored graph, query, model, prompt, and budget are held constant.*

Interpretation. The proposition is modest but foundational. A graph does not determine an answer. It defines a space of possible evidentiary contexts, and policy selects among them.

Proposition 3.2 (Recursive divergence)**.** *If policy-conditioned answers generate different candidate edits with positive probability, and governance accepts at least some policy-specific edits, then two graph processes initialized at the same $\mathcal{G}_0$ may satisfy $D_G(i, j \mid T) > 0$ for some T.*

Interpretation. Identical external memories can become different external memories without any change in the initial evidence, solely because the retrieval route altered what was proposed and subsequently validated.

3.5 Navigation elasticity

To quantify policy sensitivity, define navigation elasticity of an outcome metric Y:

$$\mathcal{E}_{Y,\pi} = \frac{\Delta Y / Y}{\Delta \pi / \pi}, \tag{18}$$

where "change in policy" must be operationalized through a parameter such as diffusion strength, exploration probability, depth limit, centrality weight, contradiction quota, or temporal decay. In empirical work, discrete policy contrasts may be more interpretable:

$$\mathrm{ATE}_{i,j} = \mathbb{E}[Y(\pi_i) - Y(\pi_j)]. \tag{19}$$

This counterfactual framing is crucial. Observing that a graph-based system performed well does not identify whether performance came from graph structure, seed selection, traversal, context packing, the model, or the governance filter.

4 A Taxonomy of Epistemic Navigation

4.1 Navigation policies as styles of inquiry

Traversal policies are often classified by algorithm. For institutional use, they should also be classified by the question they implicitly ask of the graph. Different policies encode different epistemic priorities.

Table 1. Navigation regimes and their epistemic signatures.

Policy	Implicit question	Characteristic strength	Characteristic risk
Semantic	What is most similar to the query?	Fast direct relevance over heterogeneous text	Similarity trap; weak relational reasoning
Local expansion	What surrounds the best seed within k hops?	Relational detail and explainable neighborhoods	Local confinement; seed dependence
Diffusion	What becomes important when relevance spreads through links?	Associative integration and multi-hop reach	Centrality bias; hub amplification
Community	Which modules contain the main themes?	Corpus-level sensemaking and coverage	Summary loss; community boundary errors
Typed-path	Which permitted relation sequences connect the entities?	Domain logic, causality, and auditability	Ontology dependence; missed untyped links
Goal-directed	Which next move most improves the answer?	Adaptive multi-step reasoning	Model-induced search bias; unstable traces
Contrastive	What evidence challenges the leading hypothesis?	Falsification, dissent, and risk control	Higher cost; artificial balance
Exploratory	Which weak ties or distant regions may matter?	Novel recombination and discovery	Noise; spurious distant connections
Temporal	What predecessors, updates, and outcomes define the current state?	Decision lineage and changing facts	Recency or archival bias
Decision-memory	What prior decisions resembled this one, and what followed?	Institutional learning from consequences	Analogy error; inherited policy bias

Policy	Implicit question	Characteristic strength	Characteristic risk
Hybrid router	Which policy mix fits the task and risk?	Adaptation across heterogeneous queries	Router opacity; compounded complexity

4.2 Semantic and local policies

Semantic retrieval is usually the first baseline because embeddings compress lexical and conceptual similarity into an efficient ranking. It is indispensable for seed discovery in weakly structured corpora. But similarity is not implication. A node can be semantically distant yet causally decisive; a highly similar passage can merely restate the query.

Local expansion starts from one or more seeds and follows adjacent edges up to a depth or cost limit. This makes relations visible, but the result can be dominated by seed choice. An erroneous seed creates a locally coherent but globally irrelevant context. Seed robustness should therefore be measured explicitly through repeated retrieval under alternative plausible seeds.

4.3 Diffusion and community policies

Diffusion policies propagate relevance over the graph. PageRank and its personalized variants assign importance through recursively weighted links (Brin and Page, 1998; Haveliwala, 2002). In knowledge systems, diffusion can retrieve associative chains that embedding similarity misses. However, unless degree effects are corrected, already-central nodes receive more activation, and repeated write-back around them may amplify their dominance.

Community policies exploit modularity: tightly connected regions can be summarized and selected as units (Newman, 2006; Blondel et al., 2008; Traag et al., 2019). This supports global questions, but a community is an algorithmic partition, not an ontological truth. Boundary nodes, weak ties, and minority explanations may be compressed away.

4.4 Typed, causal, and temporal paths

A typed-path policy constrains transitions to relation sequences appropriate to the task. For example:

$$\texttt{Company} \xrightarrow{\texttt{exposed\_to}} \texttt{Risk} \xrightarrow{\texttt{mitigated\_by}} \texttt{Action} \xrightarrow{\texttt{approved\_in}} \texttt{Decision}. \tag{20}$$

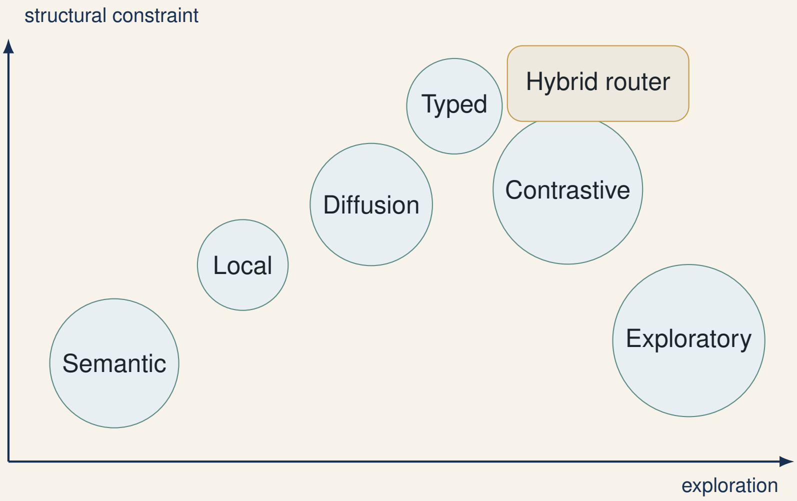
Typed traversal improves explainability and can distinguish association from causation. It should not, however, claim causal validity merely because an edge is labeled causal. Causal interpretation requires assumptions, temporal ordering, and appropriate evidence (Pearl, 2009).

Temporal traversal adds predecessor, supersession, effective date, and outcome edges. This is essential when current knowledge is contingent on earlier decisions or when the same proposition changes status over time. A system that retrieves the latest note without its decision lineage may be current but unintelligent; one that privileges history without decay may be faithful but obsolete.

4.5 Contrastive and exploratory policies

Contrastive traversal actively searches for contradiction, disconfirming evidence, competing mechanisms, and adverse outcomes. Its purpose is not to maximize diversity for its own sake, but to test whether the leading conclusion survives a structured challenge.

Exploratory traversal rewards peripheral nodes, weak ties, cross-community bridges, and long paths. It operationalizes the exploration side of the organizational learning trade-off (March, 1991). It can support recombinant discovery (Weitzman, 1998), but only if the governance layer distinguishes a useful distant connection from a merely surprising one.



Policies occupy different positions between exploitation of known relevance and exploration of distant structure. The vertical axis represents the degree to which permitted paths are constrained by ontology, type, or governance.

Figure 2. A conceptual map of navigation regimes. Positions are analytical rather than empirical estimates.

4.6 Policy composition

In practice, strong navigation is often compositional:

1. semantic retrieval identifies candidate seeds;
2. typed expansion follows decision-relevant relations;
3. temporal traversal reconstructs lineage;
4. diffusion identifies associative support;
5. contrastive search opens a dissent branch;
6. context packing balances evidence, authority, and budget.

The policy should therefore be represented as a plan, not merely a label. A "hybrid" policy without an explicit sequence, weights, quotas, and stopping rule is difficult to reproduce or audit.

5 How Navigation Reshapes the Graph

5.1 Retrieval exposure and endogenous centrality

Let $r_t(v)$ indicate whether node v is retrieved at time t, and let $w_t(v)$ denote the number or weight of accepted write-backs incident to v. Suppose:

$$\Pr(r_t(v) = 1) = g\big(\mathrm{rel}(v, q_t), d_t(v), c_t(v), \mathrm{auth}(v)\big), \tag{21}$$

where $d_t(v)$ is degree and $c_t(v)$ centrality. If retrieval increases the probability that a candidate relation is proposed around v, then:

$$\mathbb{E}[w_t(v) \mid r_t(v) = 1] > \mathbb{E}[w_t(v) \mid r_t(v) = 0]. \tag{22}$$

When the retrieval function also rewards degree or centrality, a feedback mechanism emerges:

$$d_t(v) \uparrow \Rightarrow \Pr(r_{t+1}(v) = 1) \uparrow \Rightarrow \mathbb{E}[w_{t+1}(v)] \uparrow \Rightarrow d_{t+1}(v) \uparrow. \tag{23}$$

This resembles preferential attachment, in which connectivity produces cumulative advantage (Barabási and Albert, 1999). The analogy is not exact: institutional graphs are typed, governed, query-conditioned, and subject to deletion or deprecation. But the mechanism warns that centrality can become partly endogenous to the retrieval system rather than solely a property of the underlying domain.

Proposition 5.1 (Retrieval-induced cumulative advantage)**.** *Under a policy that positively weights current degree or centrality, and a write-back process whose expected accepted incident edits increase with retrieval exposure, expected degree concentration can increase over repeated queries even when the exogenous distribution of relevant facts is stationary.*

5.2 Policy attractors and lock-in

Path dependence arises when early events alter the returns to later choices (Arthur, 1989). In a recursive knowledge system, an early retrieval decision can create a new edge; the edge can improve the future retrievability of its endpoints; later answers can then accumulate around the same region. Eventually the system may enter a policy attractor: a portion of the graph repeatedly supplies the context that justifies further growth of that portion.

Possible attractors include:

- **hub dominance**: a few entities become default bridges for many questions;
- **community enclosure**: retrieval rarely crosses established modules;
- **ontology lock-in**: typed traversal reinforces only relations already encoded in the schema;
- **recency capture**: new material crowds out enduring but older evidence;
- **decision precedent capture**: prior decisions become self-validating templates;
- **consensus compression**: contradiction nodes survive physically but disappear from active contexts.

Lock-in does not require an erroneous model. It can emerge from locally reasonable choices repeated over time.

5.3 The opposite failure: novelty inflation

The solution is not unlimited exploration. A policy that rewards distant or weakly connected nodes may produce many candidate relations that are novel but unsupported. If human reviewers face limited time, novelty can consume the validation budget and lower overall scrutiny. Moreover, once a speculative edge is admitted, it can become a bridge that makes further speculative paths appear structurally legitimate.

The design problem is therefore a governed exploration–exploitation balance. In organizational learning, too much exploitation reduces adaptation, while too much exploration dissipates effort (March, 1991). In knowledge graphs, the analogous balance concerns exploitation of high-relevance, high-authority paths versus exploration of low-centrality, cross-community paths.

5.4 Contradiction as a topological asset

Many knowledge systems treat contradiction as a data-quality defect to be resolved. Institutional reasoning often requires the opposite. Two analyses can disagree because they use different assumptions, dates, jurisdictions, or objectives. A contradiction node or edge can preserve this structure without forcing premature consensus.

A contrastive policy should therefore retrieve:

- claims that explicitly contradict the leading claim;
- alternative mechanisms leading to different outcomes;
- decisions with superficially similar facts but opposite results;
- evidence outside the dominant community;
- and sources with higher authority but lower semantic similarity.

Contradiction retention is not only a content property. It is a navigation property: dissent that is stored but never retrieved is functionally absent.

Definition 5.1 (Epistemic reachability)**.** *A node or claim is epistemically reachable for a class of queries if an admissible navigation policy retrieves it with non-negligible probability under the applicable budget and access constraints.*

Corollary 5.1. *Physical presence in the graph does not guarantee epistemic preservation. A minority view can remain stored while becoming effectively forgotten if its reachability approaches zero.*

5.5 Graph divergence as institutional divergence

Consider two teams that begin with identical vaults. Team A uses centrality-weighted diffusion; Team B uses typed paths plus a mandatory contrastive branch. After one query, their answers may differ slightly. After one hundred governed write-backs, their graphs may differ in degree concentration, contradiction density, community boundaries, and decision precedents. The teams no longer possess the same institutional memory.

This divergence is not necessarily pathological. Different objectives may justify different cognitive topologies. The governance requirement is that the divergence be visible, attributable, and periodically tested against shared tasks and shocks.

6 Governance of the Navigation Layer

6.1 The model must not become the authority

A system may be highly autonomous in search and still remain governed in knowledge authorization. The essential distinction is:

$$\text{model proposes} \neq \text{institution knows.} \tag{24}$$

Candidate knowledge should be marked as candidate knowledge. Its sources, inference class, policy, and approval status should remain attached. Governance does not mean that every low-risk edge requires a committee vote. It means that authority is explicit, proportional to risk, and separable from generation.

6.2 Why source citations are insufficient

An answer can cite every retained source and still conceal a defective search process. It may have:

- started from a biased seed;
- excluded a permitted relation type;
- stopped before reaching an adverse outcome;
- over-weighted central nodes;
- ignored a separate community;
- discarded contradictory evidence during context packing;
- or used a policy inappropriate to the question's risk.

Answer-level provenance explains what appears in the final text. Navigation provenance explains how the evidentiary set came to exist.

6.3 The Navigation Record

The framework introduces a new canonical knowledge object.

Canonical object: Navigation Record

```
object_type: navigation_record
query_id: Q047
graph_version: G_12
policy_id: contrastive_typed_v2
policy_parameters:
  semantic_seeds: 8
  maximum_depth: 4
  contradiction_quota: 0.20
  temporal_horizon_years: 5
budget:
  node_visits: 60
  source_tokens: 12000
seed_nodes: [Company_A, Interest_Rate_Shock]
edge_types_allowed:
  [exposed_to, mitigated_by, contradicts, decided_in]
communities_entered: [Real_Estate, Credit, Regulation]
visited_nodes: 43
retained_nodes: 17
rejected_branches: 6
stopping_reason: evidence_saturation
context_hash: "..."
model_id: "..."
governance_profile: high_consequence_decision
```

The record should include the version of the graph because the same policy over a different graph is not the same treatment. It should include the context hash because a visited trace does not establish which evidence the model actually received. It should include stopping because early termination can be as consequential as an incorrect transition.

6.4 Policy registry and risk tiers

Every approved navigation policy should have a registry entry:

- intended task classes;
- prohibited uses;
- eligible node and edge types;
- minimum authority and provenance requirements;
- budget bounds;
- exploration and contradiction quotas;
- stopping criteria;
- validation results;
- known failure modes;
- version and owner;
- and required human gate.

Tier	Example	Navigation requirement	Write-back authority
Low	Personal research summary	Semantic or hybrid retrieval; provenance logged	Automatic candidate note; no canonical claim
Moderate	Internal market analysis	Typed and temporal checks; contradiction sample	Analyst validation before promotion
High	Investment recommendation	Multi-policy retrieval; mandatory dissent branch; source authority thresholds	Named approver and versioned decision object
Critical	Regulatory, legal, or safety action	Independently replicated navigation; access and jurisdiction controls; complete audit bundle	Formal authorized decision process

Table 2. Illustrative risk-sensitive navigation and write-back tiers.

6.5 The governance pipeline

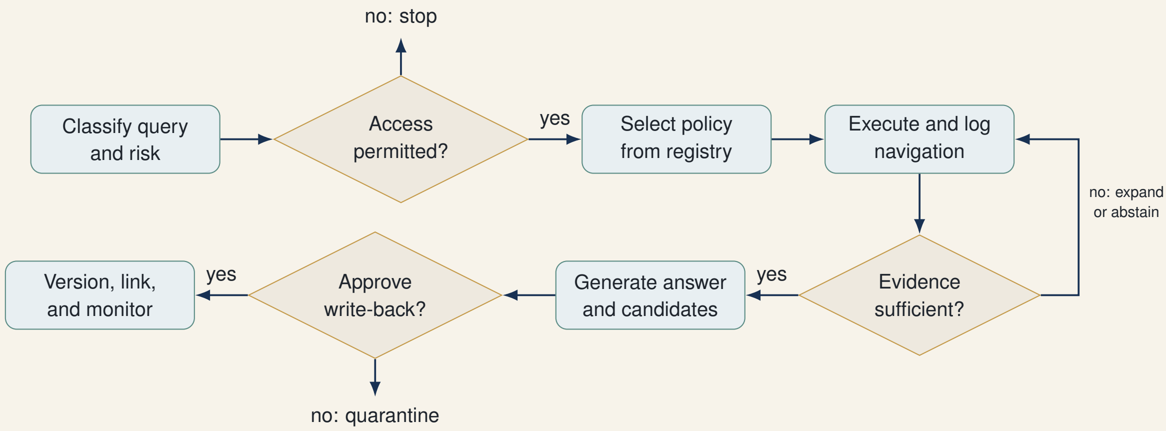


Figure 3. A governance-first navigation pipeline. Gates apply before retrieval, before generation, and before authoritative write-back.

6.6 Auditability without revealing hidden reasoning

Auditability does not require storing private chain-of-thought. The system can preserve operationally relevant artifacts:

- graph and policy versions;
- seeds, transitions, and scores;
- source identifiers and authority labels;
- visited and retained evidence;
- branch outcomes and stopping reason;
- model outputs intended for users;
- candidate knowledge objects;
- validation decisions and approver identity;
- and hashes for integrity.

This is sufficient to reproduce the retrieval environment and assess whether the process complied with policy. It is also more useful than an unstructured narrative of internal reasoning.

6.7 Periodic topology audits

Governance must inspect accumulated effects, not only individual transactions. A quarterly or threshold-triggered topology audit should test:

- degree and authority concentration;
- retrieval exposure by node type and community;
- contradiction reachability;
- orphan-node survival;
- policy-specific write-back rates;
- relation-type diversity;
- temporal coverage;
- reviewer burden and acceptance drift;
- and performance under held-out shocks.

This aligns with the broader principle that risk management should govern systems across design, deployment, monitoring, and change rather than treating validation as a one-time event (National Institute of Standards and Technology, 2023).

7 Selecting How the System Should Navigate

7.1 No universal best policy

The question "Which navigation policy is best?" is incomplete. Best for what task, under what graph quality, with what risk, budget, and loss function? A policy that maximizes short-answer accuracy may underperform on strategic sensemaking. A policy that discovers novel links may be inappropriate for a compliance determination. A policy that produces a comprehensive board briefing may be too expensive for routine triage.

Policy selection can be formulated as:

$$\pi_t^\star = \arg \max_{\pi \in \Pi_{\text{admissible}}} \mathbb{E}\left[U(\mathcal{A}_t, \Delta\mathcal{G}_t) - \lambda_1 C_t - \lambda_2 R_t - \lambda_3 L_t\right], \tag{25}$$

where U captures answer and memory value, C_t computational cost, R_t epistemic and operational risk, and L_t long-run topological distortion. The admissible set is restricted by permissions and governance.

7.2 A risk-sensitive routing matrix

Task signature	Preferred route	Reason
Direct factual lookup	Semantic plus source authority filter	Avoid unnecessary graph cost; prioritize exact evidence
Relationship explanation	Semantic seeds plus typed local expansion	Preserve interpretable paths between entities
Corpus-wide themes	Community summaries plus local verification	Combine coverage with source-level checks
Decision under uncertainty	Typed, temporal, and contrastive branches	Reconstruct precedent and challenge the leading case
Novel opportunity search	Exploratory cross-community traversal plus strict validation	Reward weak ties while containing false discovery
High-consequence action	Independent multi-policy ensemble	Reduce single-policy dependence and preserve dissent
Rapid operational triage	Budgeted hybrid router	Balance latency and task complexity

Table 3. Illustrative routing from task signature to navigation plan.

7.3 Multi-policy ensembles

For consequential questions, a single policy can be replaced by an ensemble:

1. a relevance branch retrieves the strongest direct evidence;
2. a structural branch follows typed multi-hop relations;
3. a historical branch reconstructs decisions and outcomes;
4. a dissent branch seeks contradiction and adverse cases;
5. an integrator compares claims, sources, and omissions.

The branches should remain distinguishable. Prematurely pooling all retrieved material into one context can erase the fact that one branch found a contradiction. The integrator should report agreement, disagreement, and evidence gaps, not merely produce a smoother synthesis.

Proposition 7.1 (Plural navigation value)**.** *When policy errors are imperfectly correlated and the integration rule preserves branch provenance, a multi-policy ensemble can reduce dependence on any single navigation bias. The gain is conditional on additional cost and on governance preventing low-quality branches from being mistaken for equally authoritative evidence.*

7.4 Abstention and search escalation

A mature navigation policy can decide that the graph does not support an answer. Evidence insufficiency may arise because:

- relevant nodes are inaccessible;
- paths terminate in unsupported claims;

- authoritative sources conflict;
- the query requires evidence outside the vault;
- or the budget is too small to resolve uncertainty.

The appropriate action may be to abstain, request human input, increase the budget, or authorize external search. This is bounded rationality made explicit: the system acts under constraints and should not disguise satisficing as certainty (Simon, 1955).

8 The Empirical Programme

8.1 Identification strategy

The core experiment is deliberately simple:

$$\boxed{\begin{array}{c}\text{same graph} + \text{same model} + \text{same questions}\\ +\text{same budget} + \text{same governance} + \text{different navigation}\end{array}} \tag{26}$$

The first phase freezes the graph to estimate static effects. The second phase forks identical copies and allows governed write-back to estimate dynamic effects. The third phase exposes evolved graphs to common shocks and held-out decisions. The fourth phase evaluates adaptive policy selection and governance.

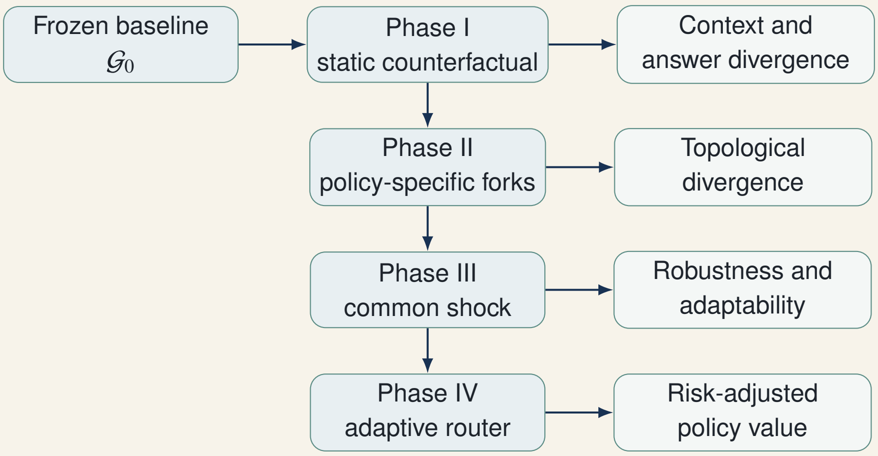


Figure 4. Controlled research design for identifying navigation effects.

8.2 Phase I: static counterfactual retrieval

Each question is executed against an unchanged graph under every policy. Random seeds are fixed or replicated. The following are recorded:

- seed nodes and seed ranks;
- visited nodes, edges, sources, and communities;
- path lengths and transition types;
- retained versus discarded evidence;
- context tokens by authority and relation type;
- complete user-facing answer;
- claim-level evidence;

- abstention, confidence, and stopping reason.

This phase estimates the effect of navigation on active context and answer without contamination from graph evolution.

8.3 Phase II: recursive governed write-back

Each policy receives a separate copy of $\mathcal{G}_0$. After every answer, candidate nodes and relations pass through the same governance protocol. Accepted changes are written only to that policy's graph:

$$\mathcal{G}_{t+1}^{(j)} = \mathcal{G}_t^{(j)} \oplus \Delta\mathcal{G}_t^{(j)}, \qquad j = 1, \ldots, J. \tag{27}$$

The same ordered question stream is applied for 25, 50, and 100 rounds. The experiment records whether divergence accelerates, stabilizes, or reverses under governance.

8.4 Phase III: shock and held-out decisions

Every evolved graph is exposed to the same new evidence and decision task. In an investment-banking laboratory, the shock might be a sharp interest-rate increase, a new regulation, a supply-chain disruption, or a change in tax treatment. The held-out task should require knowledge that was present in the initial graph but differentially reinforced or neglected during evolution.

This phase asks whether earlier navigation created:

- robust memory that adapts to a changed environment;
- brittle memory locked into a prior pattern;
- high novelty but weak authority;
- or balanced retrieval with preserved dissent.

8.5 Phase IV: navigation governance and policy selection

The final phase compares fixed policies with:

- a rule-based task router;
- a learned router trained on held-out tasks;
- a risk-sensitive router;
- and a multi-policy ensemble.

Evaluation includes not only answer quality and cost, but also long-run graph health. A router that selects the cheapest accurate policy today may still be inferior if it systematically narrows the future graph.

8.6 Hypotheses

Hypothesis 8.1 (Static context divergence). *Alternative navigation policies applied to the same graph and query will produce materially different active contexts, measured by node, edge-type, community, source-authority, and temporal overlap.*

Hypothesis 8.2 (Answer divergence). *Context divergence will translate into differences in completeness, decision recommendations, contradiction awareness, and confidence calibration, even when factual accuracy remains similar.*

Hypothesis 8.3 (Centrality amplification). *Centrality-weighted diffusion will increase retrieval exposure and accepted incident write-backs around initially central nodes, raising degree concentration relative to typed and exploration-balanced policies.*

Hypothesis 8.4 (Community enclosure). *Local and community-restricted policies will produce higher within-community density and lower cross-community bridge formation than contrastive or exploratory policies.*

Hypothesis 8.5 (Dissent preservation). *Policies with mandatory contradiction search will maintain higher epistemic reachability of dissenting claims and show lower false consensus under held-out shocks.*

Hypothesis 8.6 (Exploration cost). *Exploratory policies will generate more valid distant relations but also more rejected candidates and higher governance cost.*

Hypothesis 8.7 (Governance attenuation). *A strict, source-aware governance layer will reduce but not eliminate policy-induced graph divergence, because different policies expose reviewers to different candidate sets.*

Hypothesis 8.8 (Adaptive routing). *A task- and risk-sensitive router will dominate any single fixed policy on a composite objective combining answer value, cost, traceability, and graph health.*

8.7 Metrics

8.7.1 Navigation metrics

- node and edge visits;
- average and maximum path length;
- community coverage and transition count;
- retrieved-node centrality distribution;
- peripheral-node share;
- relation-type entropy;
- source-authority distribution;
- temporal span;
- contradiction quota achieved;
- retrieval entropy and policy overlap;
- cost, latency, and context utilization.

8.7.2 Answer metrics

- factual correctness;
- claim completeness;
- decision relevance;
- evidence sufficiency;
- traceability;
- contradiction awareness;
- novelty and non-triviality;
- calibration and abstention quality;
- recommendation stability under paraphrase.

8.7.3 Graph-evolution metrics

- graph edit distance from $\mathcal{G}_0$;
- degree Gini coefficient and centralization;
- modularity and clustering;
- connected components and orphan survival;
- cross-community bridge precision;
- relation-type diversity;
- contradiction retention and reachability;
- authoritative-to-provisional ratio;
- source lineage depth;
- policy-pair divergence $D_G(i, j \mid T)$.

8.8 Statistical analysis

Questions should be blocked by type and difficulty. Policy order should be randomized during static evaluation to control shared service drift. Multiple model samples should be collected when generation is stochastic. Mixed-effects models can separate policy, query, and model effects:

$$Y_{iqm} = \alpha + \beta_i \operatorname{Policy}_i + \gamma_q + \delta_m + u_{\text{task class}} + \varepsilon_{iqm}. \tag{28}$$

For graph evolution, trajectories are the unit of analysis. Bootstrap intervals across question sequences can estimate uncertainty in terminal graph metrics. A permutation test can assess whether observed divergence exceeds divergence induced by generation noise under a common policy.

8.9 Threats to validity

The experiment must guard against:

- graph-construction errors mistaken for navigation effects;
- model updates during the experiment;
- inconsistent token or visit budgets;
- governance reviewers learning from one policy's outputs while rating another;
- answer graders preferring verbosity or novelty;
- policy implementations differing in more than traversal;
- and synthetic results failing to transfer to real institutional graphs.

The cleanest design begins with a synthetic but realistic vault whose ground truth is known, then repeats the experiment on a de-identified real vault with expert evaluation.

9 Implementation in an Obsidian ExoBrain

9.1 The vault as a governed graph substrate

Obsidian is useful for the experiment because Markdown notes and links remain legible outside any proprietary graph database. The vault can preserve:

- stable note identifiers;
- YAML metadata;
- typed wiki-links;

- source notes and document excerpts;
- decision and outcome objects;
- candidate and authoritative status;
- graph versions and audit records.

The graph engine used by the notebooks can compile this human-readable representation into NetworkX or another graph structure for analysis, then write approved objects back as Markdown.

9.2 Recommended canonical object types

Object	Purpose	Minimum governance fields
Entity	Company, person, instrument, jurisdiction, concept	Stable ID, aliases, owner, sources
Claim	A proposition that can be supported, contradicted, or superseded	Status, confidence, source, effective dates
Evidence	Primary or secondary support for a claim	Source authority, excerpt, date, integrity hash
Decision	Authorized choice under stated assumptions	Decision maker, date, alternatives, rationale
Outcome	Observed consequence of a decision or event	Measurement method, horizon, uncertainty
Candidate relation	Machine- or human-proposed edge	Proposer, evidence, policy, review status
Navigation record	Reproducible traversal and context construction	Graph version, policy, trace, budget, hash
Policy record	Approved traversal plan and limitations	Version, task class, risk tier, validation
Audit event	Review, approval, rejection, correction, or deprecation	Actor, timestamp, reason, affected IDs

Table 4. Canonical objects for a navigation-aware ExoBrain.

9.3 The four-notebook extension

NB13 - Graph Navigation Policy Laboratory

Build a common interface for semantic, local, diffusion, community, typed, goal-directed, contrastive, exploratory, and hybrid policies. Visualize visited traces and retained contexts. Validate that budgets and access rules are comparable.

NB14 - Counterfactual Context and Answer Comparison

Freeze the graph. Execute the same question battery under every policy. Produce policy-pair overlap matrices, context maps, answer comparisons, evidence audits, and statistical estimates of static policy effects.

NB15 - Recursive Write-Back and Topological Divergence

Fork the baseline vault by policy. Run governed candidate generation and write-back over repeated rounds. Measure centrality, modularity, contradiction reachability, bridge precision, and pairwise graph divergence at checkpoints.

NB16 - Navigation Governance, Audit, and Policy Selection

Implement the policy registry, risk tiers, navigation records, gates, audit bundle, abstention rules, and adaptive router. Evaluate policy choice on held-out tasks and common shocks using a composite risk-adjusted objective.

9.4 A transparent run manifest

Every notebook run should emit a machine-readable manifest:

Minimum run manifest

```
run_id
timestamp
notebook_version
graph_version_before
graph_version_after
query_set_hash
model_id
generation_parameters
policy_id
policy_parameters
budget
access_profile
navigation_record_ids
candidate_edit_ids
governance_decisions
metric_outputs
random_seed
environment_fingerprint
```

The manifest makes the experiment reproducible and supports later forensic review. It also prevents a common failure: reporting a graph result without knowing which policy, graph version, prompt, or approval rule produced it.

9.5 A practical write-back discipline

The vault should not be overwritten in place during experimentation. The recommended structure is:

1. preserve an immutable baseline;
2. create one branch or destination vault per policy;
3. write candidates into a quarantine namespace;
4. record validation as a separate audit event;
5. promote accepted objects through versioned operations;
6. compute checksums and graph metrics at each checkpoint;
7. export a comparison bundle after every phase.

This allows any evolved graph to be traced back to the same starting point and prevents accidental contamination between treatments.

10 Institutional and Epistemological Implications

10.1 The vault is not the memory actually used

Institutions often evaluate memory systems by asking what has been stored. The navigation framework adds a second question: what is systematically retrieved? The difference resembles the distinction between an archive and a working canon. A claim can exist in the archive but remain absent from decisions because no approved policy reaches it.

This has direct implications for knowledge management. Completeness of storage does not imply completeness of use. Diversity of content does not imply diversity of active context. Preservation of dissent does not imply accessibility of dissent.

10.2 Navigation policy as institutional procedure

Organizations already recognize that procedure influences outcomes. Courts govern admissibility and discovery; scientific disciplines govern search, replication, and falsification; investment committees govern evidence, challenge, and approval. An LLM navigation policy is a computational procedure for constructing an evidentiary record.

It should therefore be treated like other consequential procedures:

- explicit rather than tacit;
- versioned rather than mutable without notice;
- appropriate to task and jurisdiction;
- monitored for disparate blind spots;
- challengeable by authorized users;
- and periodically validated against outcomes.

10.3 Memory can become performative

When the graph influences decisions and decisions are written back, the graph does not merely describe the institution. It helps constitute its future behavior. A recommendation retrieved frequently may become precedent; a relation made central may attract further analysis; a community summary may become the language through which managers interpret new evidence.

This performative character increases the importance of separating:

$$\text{frequency of retrieval} \quad \text{from} \quad \text{truth, authority, and importance.} \tag{29}$$

Centrality can be useful evidence of organizational salience, but it can also be an artifact of earlier policy choices.

10.4 Plural navigation as cognitive checks and balances

High-consequence institutions should not rely on a single route through their memory. Plural navigation creates a computational analogue of checks and balances: one branch seeks the strongest supporting case, another reconstructs causal and temporal lineage, and another searches for dissent.

The objective is not forced symmetry. Evidence can overwhelmingly support one conclusion. The objective is to ensure that the conclusion survives more than one epistemic route and that disagreement remains visible when it matters.

10.5 From explainable answers to explainable access

Explainable AI discussions often focus on why a model produced a prediction. In retrieval-augmented institutional systems, explanation must begin earlier:

> Why was this evidence available to the model, while other evidence was not?

This is explainable access. It includes seed choice, path permissions, rank functions, budgets, and stopping. Without it, the organization may explain the final synthesis while leaving the selection mechanism opaque.

11 Limitations and Research Agenda

11.1 Limits of the present theory

The framework is intentionally broad and therefore leaves several mechanisms to empirical specification.

First, graph topology is only one representation of organizational knowledge. Important evidence may remain in tables, documents, images, code, or human expertise. A navigation policy that is excellent over the graph may still fail because graph construction omitted critical material.

Second, policy effects interact with model behavior. A model may ignore retrieved evidence, over-weight its first items, or synthesize conflicting sources differently. Holding the model constant identifies policy effects only for that model and prompting regime.

Third, governance is not an exogenous perfect filter. Reviewers have limited time, may be influenced by presentation order, and may disagree about authority. The governance layer can attenuate, transmit, or amplify navigation bias.

Fourth, graph metrics are not direct measures of knowledge quality. Higher diversity, lower centralization, or more cross-community edges are not automatically better. Metrics require task- and domain-specific interpretation.

Fifth, a navigation trace can be reproducible while the underlying policy remains strategically misaligned. Auditability is necessary, not sufficient.

11.2 Research questions

The theory motivates a broader agenda:

1. How stable are navigation effects across models, prompts, and graph construction methods?
2. Which policy dimensions drive the largest static and dynamic divergence?
3. Can long-run topological distortion be predicted from short-run retrieval exposure?
4. What contradiction quota improves robustness without producing false balance?
5. How should exploration budgets vary with decision risk and graph maturity?
6. Can an adaptive router be constrained by auditable rules while retaining learning benefits?
7. How do human reviewers react to policy-labeled versus policy-blind candidate edits?

8. When should a system repair the graph, and when should it preserve ambiguity?
9. How can cross-vault or multi-institution navigation preserve access boundaries?
10. What constitutes a legally or scientifically sufficient navigation record?

11.3 Three maturity transitions

The ExoBrain research programme can now be understood as three transitions.

Transition	Central question	Principal control
Induced Structure	How can LLM interaction reveal and propose relationships in a proprietary corpus?	Provenance and candidate status
Governed Knowledge Evolution	How can validated knowledge enter memory without making the model the authority?	Approval gates, versions, and canonical objects
Navigation-Conditioned Evolution	How does the route through memory shape present reasoning and future topology?	Policy registry, navigation records, counterfactual tests, and topology audits

Table 5. The progression from graph construction to governed cognitive topology.

12 Conclusion

Building a knowledge graph is not the end of the external-memory problem. It creates a structured possibility space. The model still requires a method for moving through that space, and every method privileges some evidence, some relations, some communities, and some histories over others.

The paper has proposed a simple but consequential formula:

$$\boxed{\text{institutional cognition} = \text{knowledge topology} \times \text{navigation policy} \times \text{governance}} \tag{30}$$

The multiplicative notation is deliberate. A rich graph with a defective navigation policy can produce narrow context. Sophisticated navigation without governance can turn model inference into unauthorized memory. Strong governance over an impoverished graph can preserve integrity while failing to produce insight.

At the static level, navigation conditions the evidence from which an answer is generated. At the dynamic level, it conditions the candidate knowledge that may enter the next graph. Repetition turns retrieval into an evolutionary force. Central nodes may become more central; communities may become more enclosed; contradictions may become unreachable; exploratory bridges may create discovery or noise. Two identical vaults can evolve apart because they were traversed differently.

The practical response is not to search for a universally optimal algorithm. It is to make navigation explicit, plural, risk-sensitive, reproducible, and governable. The organization needs a policy registry, task routing, contradiction and exploration controls, navigation records, versioned write-back, and periodic audits of accumulated topology. For high-consequence questions, it needs more than one path.

The research programme is empirically tractable. Freeze one graph; hold model, prompt, questions, budget, and governance constant; vary navigation. Then fork the graph, permit controlled write-back, and measure divergence under common shocks. The notebooks NB13–NB16 turn the theoretical proposition into a falsifiable laboratory.

The resulting question is larger than GraphRAG performance. It concerns the architecture of institutional intelligence. The organization must know not only what its external memory contains, but how its autonomous systems reach that memory, what they systematically fail to see, and how yesterday's route becomes part of tomorrow's map.

Appendix A: Compact Mathematical Summary

$$\text{Stored graph:} \quad \mathcal{G}_t = (V_t, E_t, \tau_V, \tau_E, X_t, S_t, \Gamma_t) \tag{31}$$

$$\text{Navigation trace:} \quad \mathcal{V}_t^\pi = \mathrm{Walk}(\mathcal{G}_t, q_t; \pi_t, B_t, \Omega_t) \tag{32}$$

$$\text{Active context:} \quad C_t^\pi = P(\mathcal{V}_t^\pi, q_t, B_t) \tag{33}$$

$$\text{Answer:} \quad \mathcal{A}_t^\pi \sim \mathcal{L}(\cdot \mid q_t, C_t^\pi, \theta, p_t) \tag{34}$$

$$\text{Candidate edits:} \quad \Delta\widetilde{\mathcal{G}}_t^\pi = X(\mathcal{A}_t^\pi, C_t^\pi, \mathcal{V}_t^\pi) \tag{35}$$

$$\text{Governed edits:} \quad \Delta\mathcal{G}_t^\pi = \mathcal{H}_t(\Delta\widetilde{\mathcal{G}}_t^\pi) \tag{36}$$

$$\text{Graph transition:} \quad \mathcal{G}_{t+1}^\pi = \mathcal{G}_t^\pi \oplus \Delta\mathcal{G}_t^\pi \tag{37}$$

$$\text{Policy divergence:} \quad D_G(i, j \mid T) = d_G(\mathcal{G}_T^{\pi_i}, \mathcal{G}_T^{\pi_j}). \tag{38}$$

The central empirical estimand

For a fixed baseline graph and controlled query process, estimate the effect of assigning navigation policy π_i rather than π_j on active context, answer quality, governance burden, and terminal graph topology.

Appendix B: Minimum Audit Bundle

1. immutable baseline graph or content-addressed snapshot;
2. graph schema, ontology, and permissions profile;
3. query set and query-set hash;
4. policy registry entries and parameters;
5. navigation records for every run;
6. exact retained contexts or integrity-preserving references;
7. user-facing answers and claim-evidence maps;
8. candidate graph edits;
9. governance decisions, approvers, and reasons;
10. before-and-after graph versions;
11. metric definitions and computed results;
12. environment and model identifiers;
13. exceptions, abstentions, failures, and manual interventions;

14. reproducibility manifest.

References

Arthur, W. B. (1989). Competing technologies, increasing returns, and lock-in by historical events. *The Economic Journal*, 99(394):116–131.

Barabási, A.-L. and Albert, R. (1999). Emergence of scaling in random networks. *Science*, 286(5439):509–512.

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., and Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10):P10008.

Brin, S. and Page, L. (1998). The anatomy of a large-scale hypertextual web search engine. *Computer Networks and ISDN Systems*, 30(1–7):107–117.

Dai, J., Wu, D., Chen, Y., Zeng, Z., Yan, Y., Liu, Z., and Sun, M. (2026). Navirag: Towards active knowledge navigation for retrieval-augmented generation. *arXiv preprint arXiv:2604.12766*.

Dong, S., Zhang, Q., Xiao, Y., Chen, S., Zhou, C., and Huang, X. (2026). Use graph when it needs: Efficiently and adaptively integrating retrieval-augmented generation with graphs. *arXiv preprint arXiv:2602.03578*.

Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., Metropolitansky, D., Ness, R. O., and Larson, J. (2024). From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*.

Gutiérrez, B. J., Shu, Y., Gu, Y., Yasunaga, M., and Su, Y. (2024). Hipporag: Neurobiologically inspired long-term memory for large language models. In *Advances in Neural Information Processing Systems*, volume 37, pages 59532–59569.

Gutiérrez, B. J., Shu, Y., Qi, W., Zhou, S., and Su, Y. (2025). From rag to memory: Non-parametric continual learning for large language models. *arXiv preprint arXiv:2502.14802*.

Haveliwala, T. H. (2002). Topic-sensitive pagerank. *Proceedings of the Eleventh International Conference on World Wide Web*, pages 517–526.

He, X., Tian, Y., Sun, Y., Chawla, N. V., Laurent, T., LeCun, Y., Bresson, X., and Hooi, B. (2024). G-retriever: Retrieval-augmented generation for textual graph understanding and question answering. In *Advances in Neural Information Processing Systems*, volume 37, pages 132876–132907.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Kuttler, H., Lewis, M., Yih, W.-t., Rocktaschel, T., Riedel, S., and Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.

Li, J., Sun, Z., Zhou, Z., Qiu, S., Huang, J., Sun, H., and Qiu, L. (2025). Agentic-kgr: Co-evolutionary knowledge graph construction through multi-agent reinforcement learning. *arXiv preprint arXiv:2510.09156*.

Lin, J., Jiang, C., Lin, X., Zhang, R., Zhu, X., Liu, J., Tang, C., Du, Y., Gao, S., Ning, J., Liu, L., Huang, Z., Li, T., Ye, J., and He, J. (2026). Evograph-r1: Self-evolving multimodal knowledge hypergraphs for agentic retrieval. *arXiv preprint arXiv:2607.12764*.

Liu, R., Jiang, H., Yan, X., Tang, B., and Li, J. (2025). Polyg: Effective and efficient graphrag with adaptive graph traversal. *arXiv preprint arXiv:2504.02112*.

March, J. G. (1991). Exploration and exploitation in organizational learning. *Organization Science*, 2(1):71–87.

National Institute of Standards and Technology (2023). Artificial intelligence risk management framework (ai rmf 1.0). Technical Report NIST AI 100-1, U.S. Department of Commerce.

Newman, M. E. J. (2006). Modularity and community structure in networks. *Proceedings of the National Academy of Sciences*, 103(23):8577–8582.

Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*. Cambridge University Press, Cambridge, 2 edition.

Pirolli, P. and Card, S. (1999). Information foraging. *Psychological Review*, 106(4):643–675.

Simon, H. A. (1955). A behavioral model of rational choice. *The Quarterly Journal of Economics*, 69(1):99–118.

Sun, J., Xu, C., Tang, L., Wang, S., Lin, C., Gong, Y., Ni, L. M., Shum, H.-Y., and Guo, J. (2023). Think-on-graph: Deep and responsible reasoning of large language model on knowledge graph. *arXiv preprint arXiv:2307.07697*.

Traag, V. A., Waltman, L., and van Eck, N. J. (2019). From louvain to leiden: Guaranteeing well-connected communities. *Scientific Reports*, 9:5233.

Weitzman, M. L. (1998). Recombinant growth. *The Quarterly Journal of Economics*, 113(2):331–360.